SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR.



Predictive Analytics

Project Report

"Predicting EV Charging Station Demand using the KDD Process"

CA3 – End-to-End Predictive Model Development

COMPUTER SCIENCE AND ENGINEERING
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Abstract

Electric vehicle (EV) adoption is growing rapidly, and charging operators need to know how busy a new station will be before investing in it. This project applies the Knowledge Discovery in Databases (KDD) process to predict the average number of daily users at an EV charging station from its configuration, pricing and location attributes. A public dataset of 5,000 charging stations was selected, checked for quality, explored, transformed, and used to train four regression models: Linear Regression, K-Nearest Neighbours, Decision Tree and Random Forest. These were compared against a baseline that always predicts the mean. Random Forest achieved the lowest error (RMSE = 26.42 users/day), but no model meaningfully outperformed the baseline (RMSE = 26.43); every coefficient of determination was approximately zero. Data quality checks explain this result: the dataset shows strong signs of being synthetically generated, with physically impossible charger/capacity combinations and identical prices across charger types. The main contribution of the project is therefore a demonstration that the KDD process can reveal when data does not contain the knowledge being sought, and that better real-world data, rather than a more complex algorithm, is needed to predict charging demand reliably.

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Chapter 1

Introduction

1.1 Background

The transport sector is moving towards electrification. The International Energy Agency reports continued strong growth in global electric car sales, which increases the need for public charging infrastructure [13]. Building a charging station requires significant investment in hardware, grid connection and land. A station built in the wrong place, or with the wrong configuration, may remain under-used, while a popular station without enough capacity causes queues and driver frustration.

Predictive analytics can support these decisions by estimating the demand a station is likely to attract from its characteristics. Such an estimate helps operators prioritise locations, choose charger types and capacity, and set prices.

1.2 Problem Statement

> *Can the average number of daily users at an EV charging station be predicted from its location, technical configuration, pricing and service attributes?*

The target variable is `Usage Stats (avg users/day)`, a continuous value, so the task is treated as a **supervised regression** problem.

1.3 Objectives

1. To understand the structure and quality of a public EV charging station dataset.
2. To pre-process and transform the data into a form suitable for machine learning.
3. To build and compare several regression models for predicting daily station usage.
4. To evaluate the models against a simple baseline using R^2, MAE and RMSE.
5. To interpret the results and identify which factors, if any, drive charging demand.

1.4 Scope

The project uses one static dataset of 5,000 stations and four classical machine-learning algorithms. Time-series forecasting of hourly or daily load is outside its scope, because the dataset has no time dimension.

1.5 Organisation of the Report

Chapter 2 reviews the KDD process, the algorithms used, and earlier work on EV charging demand prediction. Chapter 3 describes each stage of the predictive modelling workflow applied in this project. Chapter 4 presents and discusses the results. Chapter 5 concludes the report and suggests future work.

Chapter 2

Literature Review

2.1 The KDD Process

Fayyad, Piatetsky-Shapiro and Smyth define Knowledge Discovery in Databases as "the non-trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data" [1]. They describe it as an iterative sequence of stages: *selection*, *pre-processing*, *transformation*, *data mining*, and *interpretation/evaluation*. They stress that data mining is only one step, and that the stages before and after it largely determine whether useful knowledge is found. CRISP-DM, a widely used industry methodology, follows a similar cycle of business understanding, data understanding, data preparation, modelling, evaluation and deployment [2]. Han, Kamber and Pei note that data pre-processing often takes most of the effort in a data mining project, because real-world data is frequently incomplete, noisy and inconsistent [3].

2.2 Data Quality

The quality of the input data limits the quality of any model. Sambasivan et al. show that undervaluing data work leads to "data cascades": problems in the data that go unnoticed and cause failures later in the pipeline [4]. This motivates the validity checks performed in this project, which go beyond looking for missing values.

2.3 Regression Algorithms

- **Linear Regression** models the target as a weighted sum of the inputs. It is fast and easy to interpret, but can only capture linear relationships [5].
- **K-Nearest Neighbours (KNN)** predicts a value by averaging the targets of the k most similar training examples. It was introduced by Cover and Hart [6]. Because it relies on distances, features must be scaled.

- **Decision Trees** (CART) recursively split the data into regions with similar target values, producing interpretable if/else rules [7]. Deep trees can overfit.
- **Random Forest** combines many decision trees, each trained on a bootstrap sample with random subsets of features. Averaging their predictions reduces variance and usually improves accuracy [8].

Kuhn and Johnson recommend comparing every model against a simple baseline and evaluating it on held-out data, so that apparent performance is not just the result of overfitting [9].

2.4 EV Charging Demand Prediction

Several studies have applied machine learning to EV charging. Majidpour et al. forecast charging load using either customer profiles or station measurements, and compared algorithms including KNN and ARIMA [10]. Straka et al. predicted the popularity of charging stations in the Netherlands, combining station data with geographic and demographic features of the surrounding area [11]. Lee, Li and Low released ACN-Data, a large open dataset of real charging sessions, to support research on charging behaviour [12].

A common finding of this work is that demand depends strongly on *context*: the surrounding population, points of interest, traffic and the time of day. Station hardware attributes alone are not enough. This is relevant to the interpretation of the results in Chapter 4.

2.5 Research Gap

Earlier studies mostly rely on real session data enriched with external context. This project examines what can be learned from station-level attributes alone in a publicly available dataset, and uses the KDD process to assess whether that data is suitable for the prediction task.

Chapter 3

Workflow / Predictive Modelling Process

Figure 3.1 shows the workflow followed, based on the KDD stages of Fayyad et al. [1]. The implementation was written in Python using pandas [16], NumPy [17], scikit-learn [15] and Matplotlib [18].

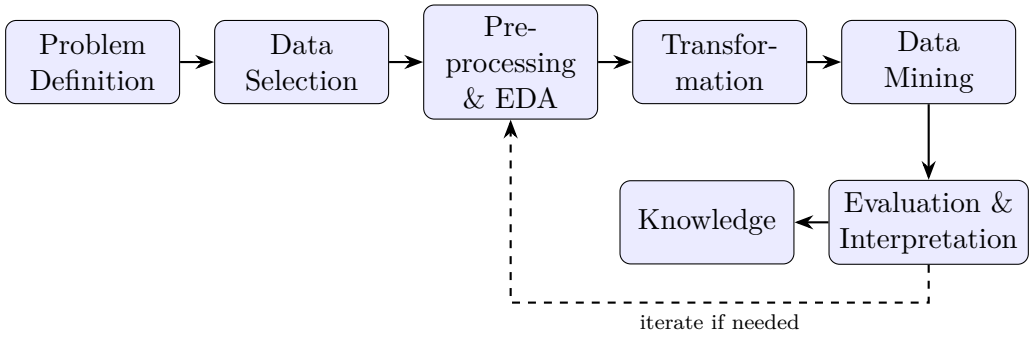


Figure 3.1: Predictive modelling workflow based on the KDD process

3.1 Stage 1: Data Selection

The *Detailed EV Charging Stations* dataset from Kaggle [14] was selected because it contains a demand measure (daily users) together with the decisions an operator controls when opening a station. It has 5,000 rows (stations) and 17 columns, summarised in Table 3.1.

Table 3.1: Dataset attributes

Attribute	Type	Description
Station ID	Identifier	Unique station code
Latitude, Longitude	Numeric	GPS coordinates
Address	Text	Street and city
Charger Type	Categorical	AC Level 1, AC Level 2, DC Fast Charger
Cost (USD/kWh)	Numeric	Charging price (0.10–0.50)
Availability	Categorical	24/7, 6:00–22:00, 9:00–18:00
Distance to City (km)	Numeric	Distance from city centre (0.5–20)
Usage Stats (avg users/day)	Numeric	**Target**: average daily users (10–100)
Station Operator	Categorical	Tesla, ChargePoint, EVgo, Greenlots, Ionity
Charging Capacity (kW)	Numeric	22, 50, 150 or 350 kW
Connector Types	Categorical	e.g. CCS, CHAdeMO, Type 2, Tesla
Installation Year	Numeric	2010–2023
Renewable Energy Source	Binary	Yes / No
Reviews (Rating)	Numeric	Average rating (3.0–5.0)
Parking Spots	Numeric	Number of bays (1–10)
Maintenance Frequency	Categorical	Monthly, Quarterly, Annually

3.2 Stage 2: Data Pre-processing

3.2.1 Missing Values and Duplicates

The dataset was checked with `isnull().sum()` and `duplicated().sum()`. It contains **no missing values** and **no duplicate rows**, so no imputation or removal was needed.

3.2.2 Outlier Detection

Box plots of the numeric attributes (Figure 3.2) show that all values lie within sensible, bounded ranges, with no extreme outliers. All records were therefore kept.

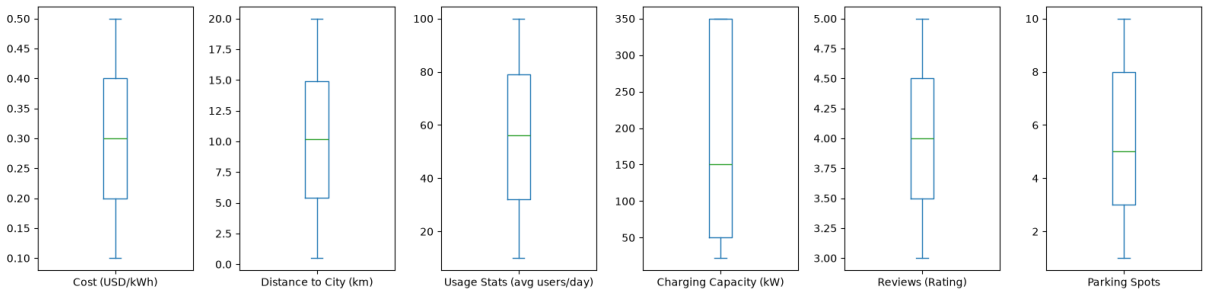


Figure 3.2: Box plots of the numeric attributes

3.2.3 Data Quality (Validity) Check

Data can be complete and still be invalid, so the attributes were checked against domain knowledge. In practice, AC Level 1 chargers deliver only about 2 kW, and DC fast chargers are more expensive than AC chargers. Table 3.2 shows that every charger type appears with every capacity in almost equal numbers, including the physically impossible combination of AC Level 1 at 350 kW. Table 3.3 shows that the average cost is identical for every charger type. In addition, 500 records (10%) have placeholder addresses such as "Random Rd, City 98, Country".

Table 3.2: Charger type vs charging capacity (number of stations)

Charger Type	**22 kW**	**50 kW**	**150 kW**	**350 kW**
AC Level 1	401	384	418	435
AC Level 2	406	479	431	448
DC Fast Charger	398	393	434	373

Table 3.3: Average cost by charger type

Charger Type	**Mean Cost (USD/kWh)**
AC Level 1	0.300
AC Level 2	0.299
DC Fast Charger	0.302

These findings suggest that the dataset is **synthetically generated**, with each attribute drawn independently at random. This was recorded as a key risk for the modelling stage.

3.2.4 Exploratory Data Analysis

The distribution of the target (Figure 3.3) is roughly uniform between 10 and 100 users per day, with a mean of about 55.

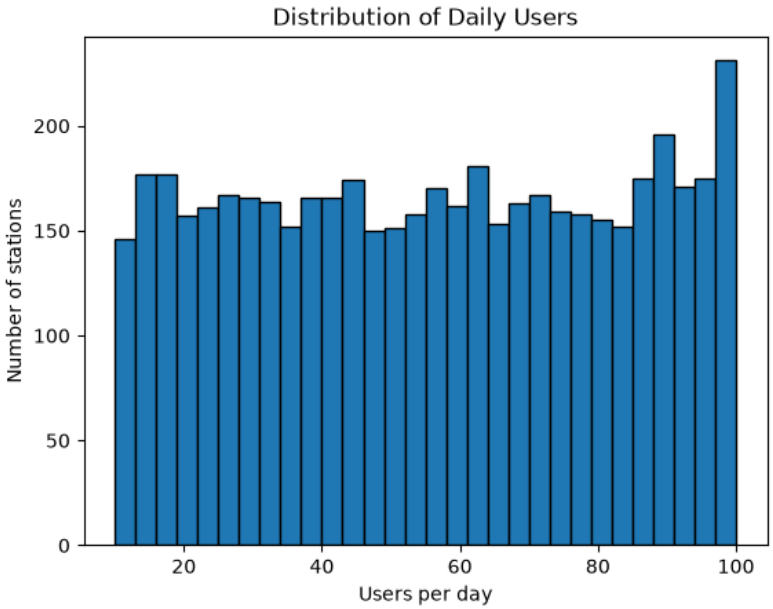


Figure 3.3: Distribution of the target variable (average users per day)

The correlation heatmap (Figure 3.4) shows that every numeric feature has a correlation with the target close to zero ($|r| < 0.035$). The features are also almost uncorrelated with each other.



Figure 3.4: Correlation heatmap of the numeric attributes

Figure 3.5 shows that average usage is almost the same for every charger type, operator and opening-hours option. Across these categories, the gap between the highest and lowest

group means is under 3 users per day.

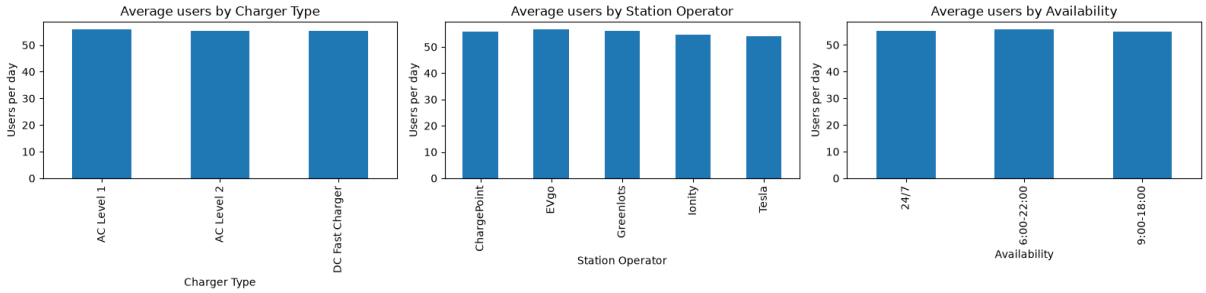


Figure 3.5: Average daily users by charger type, operator and availability

3.3 Stage 3: Data Transformation

The following transformations were applied:

1. **Feature removal:** `Station ID` and `Address` (unique identifiers with no predictive meaning) and `Latitude`/`Longitude` were dropped.
2. **Feature engineering:** a new feature, *Station Age* = 2025 − Installation Year, replaced the installation year.
3. **Encoding:** the categorical columns (charger type, availability, operator, connector types, renewable source, maintenance frequency) were converted to numbers with one-hot encoding (`pd.get_dummies`, `drop_first=True`). This produced 32 input features.
4. **Train/test split:** the data was split into 80% training (4,000 stations) and 20% testing (1,000 stations), with `random_state=42` for reproducibility.
5. **Scaling:** `StandardScaler` was fitted on the training data only and then applied to both sets. This puts all features on the same scale, which KNN requires, and avoids leaking test information into training.

3.4 Stage 4: Data Mining (Modelling)

Four regression algorithms were trained on the training set (Table 3.4). A **baseline** model that always predicts the training-set mean was used as the minimum standard: a useful model must beat it.

Table 3.4: Models and settings

Model	**Settings**
Baseline	Predict mean of training target
Linear Regression	Default
K-Nearest Neighbours	$k = 15$
Decision Tree	Maximum depth = 5
Random Forest	100 trees, maximum depth = 8

3.5 Stage 5: Evaluation

The models were evaluated on the unseen test set using three metrics, where y_i is the actual value, $\hat{y}_i$ the prediction and $\bar{y}$ the mean:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \qquad \text{MAE} = \frac{1}{n}\sum |y_i - \hat{y}_i| \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

An R^2 of 1 means a perfect fit, and an R^2 of 0 means the model is no better than predicting the mean. MAE and RMSE measure the average error in users per day; lower is better.

Chapter 4

Results and Discussion

4.1 Model Performance

Table 4.1 and Figure 4.1 show the performance of each model on the test set.

Table 4.1: Model performance on the test set (1,000 stations)

Model	R^2	MAE	RMSE
Baseline (mean)	−0.0012	22.81	26.43
Linear Regression	−0.0016	22.81	26.44
K-Nearest Neighbours	−0.0402	23.03	26.94
Decision Tree	−0.0335	22.95	26.85
Random Forest	**−0.0003**	**22.75**	**26.42**

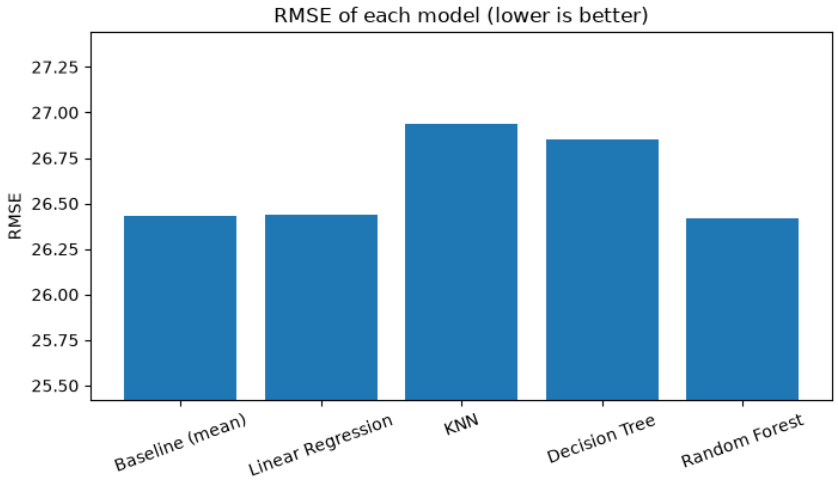


Figure 4.1: RMSE of each model (lower is better)

Random Forest achieved the lowest MAE and RMSE, but its improvement over the baseline is negligible: about 0.01 users per day in RMSE, against an average error of about 26 users per day. Linear Regression performed the same as the baseline. KNN and

the Decision Tree performed slightly *worse* than the baseline, which indicates they fitted noise in the training data. All R^2 values are approximately zero or slightly negative.

4.2 Actual vs Predicted Values

Figure 4.2 plots the Random Forest predictions against the actual values. If the model were accurate, the points would lie along the red diagonal. Instead, the predictions form a narrow horizontal band around 55 users per day, whatever the actual value. The model has effectively learned to predict the average.

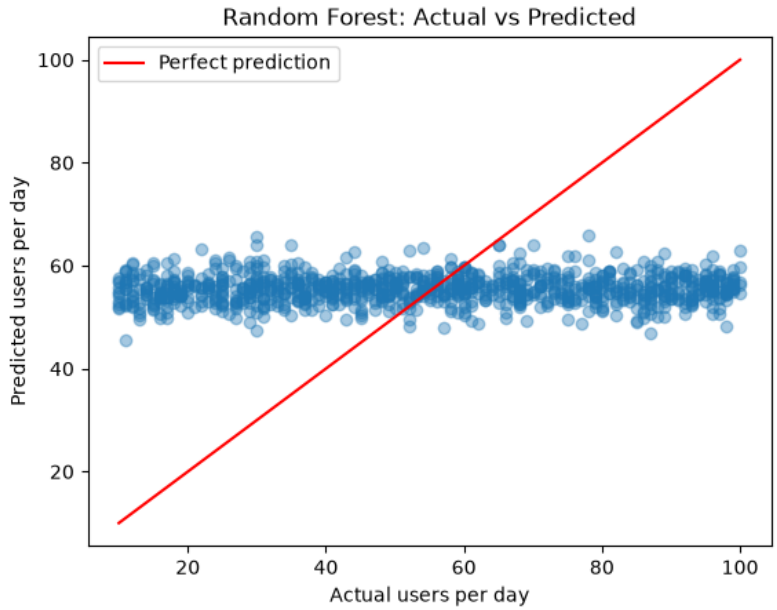


Figure 4.2: Random Forest: actual vs predicted daily users

4.3 Feature Importance

Figure 4.3 shows the ten most important features according to the Random Forest. Distance to city, cost and rating rank highest. However, these importances must be read with care. Because the model performs no better than the baseline, the ranking does not show real demand drivers. Tree-based importance favours continuous features because they offer more possible split points [5], so it reflects which columns the trees split on most often.

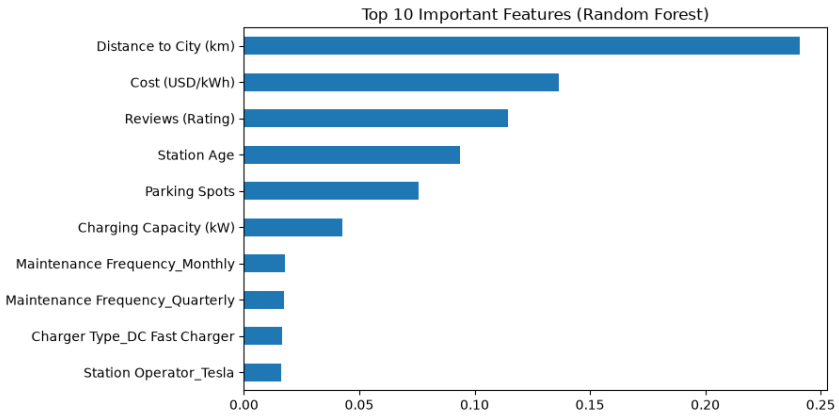


Figure 4.3: Top 10 features by Random Forest importance

4.4 Discussion

The results consistently show that the station attributes in this dataset do not predict daily usage. Several pieces of evidence point to the same cause:

- **EDA:** near-zero correlations and almost identical group means (Section 3.2.4).
- **Modelling:** four very different algorithms, linear and non-linear, all fail to beat the mean.
- **Validity check:** physically impossible and perfectly balanced attribute combinations (Section 3.2.3) indicate that the attributes were generated independently of one another and of the target.

If the target is independent of the features, no algorithm can learn a relationship, however complex it is. The weak result is therefore a property of the *data*, not a failure of the method. This matches the literature: studies with real data [11, 10] find that charging demand depends heavily on external context such as surrounding population, amenities and time, and this dataset contains no such information.

This outcome shows the value of the KDD process. The data quality check in the preprocessing stage identified the problem before any model was built, and the evaluation stage, by comparing against a baseline, prevented an over-optimistic conclusion. Reporting the Random Forest as "the best model" without the baseline comparison would have been misleading.

4.5 Limitations

- The dataset appears synthetic, so the results do not describe real-world charging behaviour.

- The target is a single static average with no time dimension, which rules out analysis of trends, seasons or peak hours.
- Important external factors, such as local EV ownership, traffic, population density and nearby amenities, are not available.
- Only four algorithms with fixed settings were tested. However, because there is no signal in the data, tuning would not be expected to change the conclusion.

Chapter 5

Conclusion

This project applied the complete KDD process (data selection, pre-processing, transformation, data mining and evaluation) to predict the daily usage of EV charging stations. The data was clean in terms of missing values, duplicates and outliers. Four regression models were trained and compared against a mean baseline.

The best model, Random Forest, achieved an RMSE of 26.42 users per day, compared with 26.43 for the baseline, and an R^2 of approximately zero. None of the models could predict station usage in a meaningful way. Data quality analysis showed that the dataset is very likely synthetically generated, with attributes unrelated to each other and to the target.

The key knowledge discovered is therefore that **a predictive model can only be as good as its data**. The KDD process worked as intended: it detected that the data does not contain the information needed to answer the business question, and it prevented a misleading model from being presented as useful. For a charging operator, the recommendation is to collect real demand data before relying on predictive models for investment decisions.

Future Work

1. Apply the same workflow to real charging-session data, such as ACN-Data [12] or open city charging datasets.
2. Add external features such as population density, EV registrations, traffic volume and nearby points of interest, following Straka et al. [11].
3. Use time-series methods to forecast demand by hour and day once time-stamped data is available.
4. Apply cross-validation and hyperparameter tuning to compare the models more robustly.

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